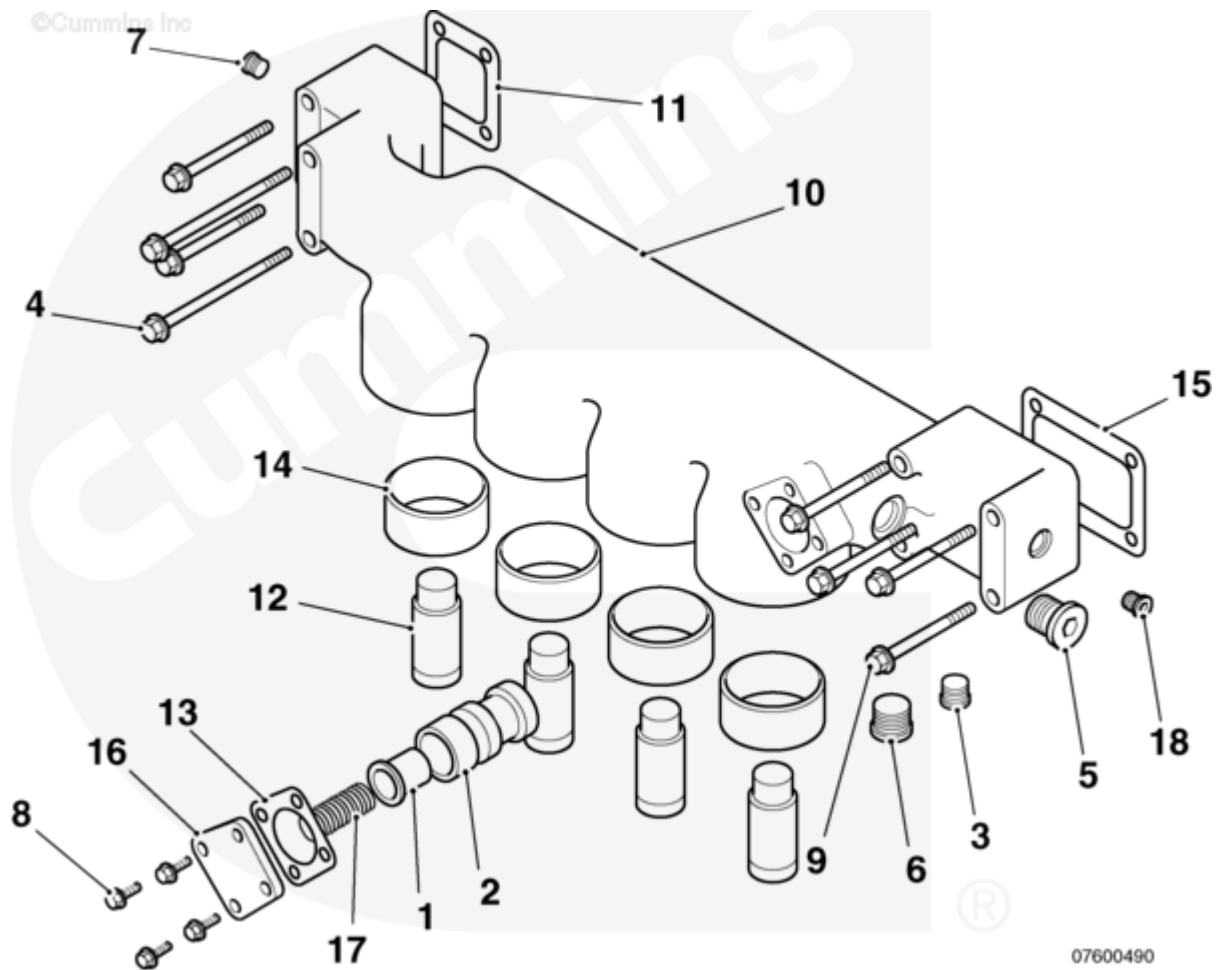


Exploded View



One Piece Oil Filter Head

1. Pressure regulator plunger
2. Bypass valve body
3. Threaded plug
4. Hexagon heap capscrew
5. Threaded plug
6. Threaded plug
7. Threaded plug
8. Hexagon head capscrew
9. Hexagon head capscrew
10. Lubricating oil filter head
11. Filter head gasket
12. Filter head adapter
13. Cover plate gasket

14. Filter head adapter
15. Filter head gasket
16. Cover plate
17. Compression spring
18. Threaded plug.

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

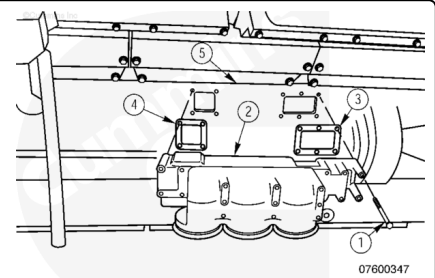
- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html)

Remove

Two-Piece Filter Head, Spin-on Filter

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use guide pins for removal and installation of the lubricating oil filter head.

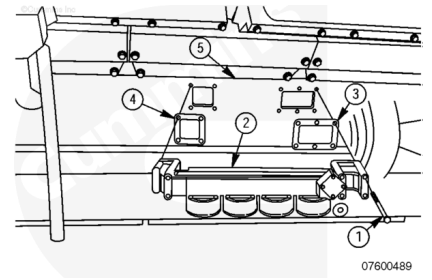
If the engine is equipped with pre-oil filter and post-oil filter sensors, remove the sensors.

Remove the 12 capscrews (1), filter head (2), and two gaskets (3 and 4) from the engine (5).

One-Piece Filter Head, Spin-on Filter

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use guide pins for removal and installation of the lubricating oil filter head.

If the engine is equipped with pre-oil filter and post-oil filter sensors, remove the sensors.

Remove the eight capscrews (1), filter head (2), and two gaskets (3 and 4) from the engine (5).

Disassemble

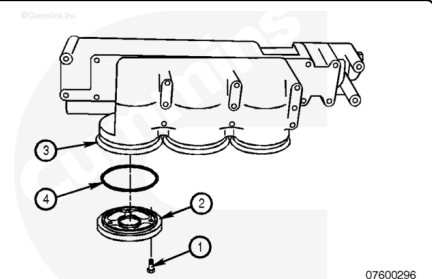
Two-Piece Filter Head, Spin-on Filter

Lubricating oil filter heads that use the combination oil filter have an adapter for the filter.

Remove the four adapter capscrews (1).

Remove the filter head adapter (2) from the oil filter head (3).

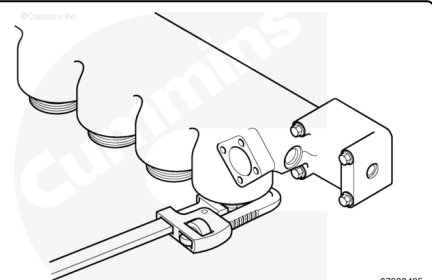
Remove and discard the o-ring (4).



One-Piece Filter Head, Spin-on Filter

Remove the adapter **only** if the threads are damaged. Removing the adapter will cause damage to the threads.

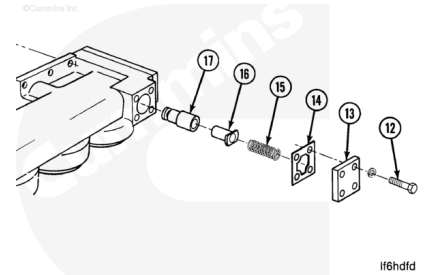
Use a large pipe wrench to remove the adapter from the filter head.



Remove the four capscrews (12) and the lock washers.

Remove the following:

- Cover plate (13)
- Gasket (14)
- Filter bypass valve spring (15)
- Plunger (16)
- Housing (17).

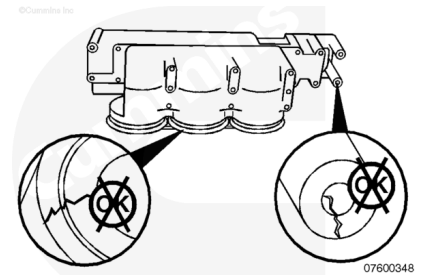


Clean and Inspect for Reuse

Two-Piece Filter Head, Spin-on Filter

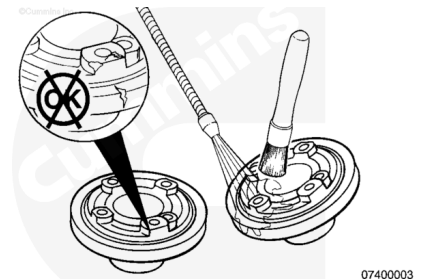
Inspect the lubricating oil filter head for cracks or other damage.

If any of the components are damaged, the filter head **must** be rebuilt or replaced.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the adapter with solvent and dry with compressed air.

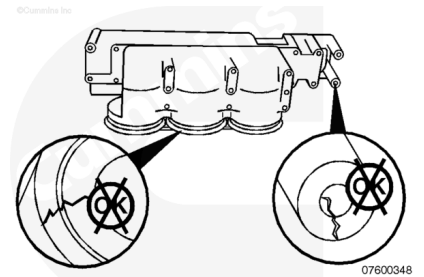
Inspect the adapter for cracks and other damage.

If any damage is found, the filter head **must** be replaced.

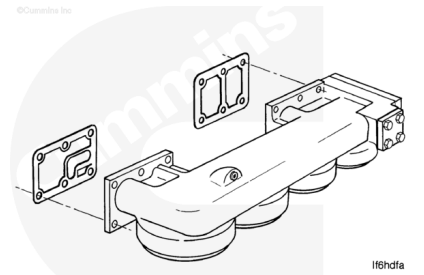
One-Piece Filter Head, Spin-on Filter

Inspect the lubricating oil filter head for cracks or other damage.

If any of the components are damaged, the filter head **must** be rebuilt or replaced.

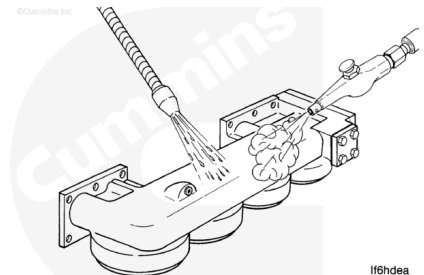


Use a gasket scraper to remove the two mounting gaskets if they remained on the filter head.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

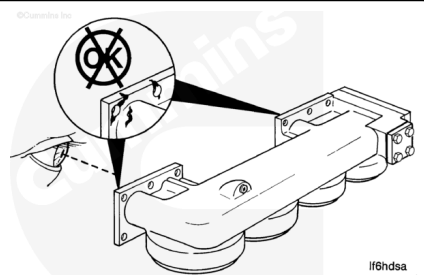
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean all gasket and sealing surfaces.

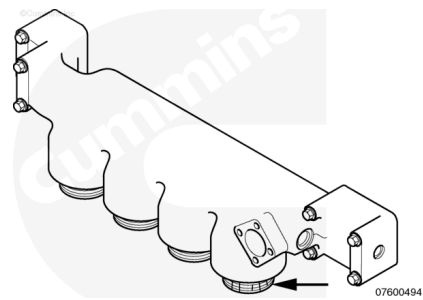
Use solvent. Clean the oil filter head.

Dry with compressed air.

Inspect the oil filter head for cracks or other damage.

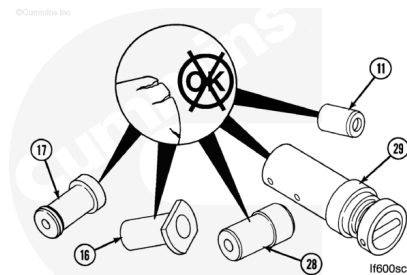


Inspect the oil filter adapter threads for signs of damage.
Replace the adapter if the threads are damaged.



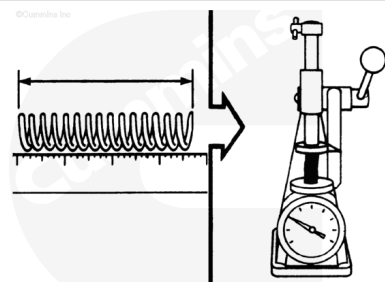
Inspect the surfaces of all the plungers (16, 17, 28, 29) and their mating bores (11) in the housing, or the filter head.

If it is **not** possible to remove the scratches or marks with a fine crocus cloth, the part **must** be replaced.



Measure the free length of the spring.

Free Length		
mm		in
93.65	MIN	3.687
96.85	MAX	3.813



Compress the spring using the tool to a height of 69.85 mm [2.75 in].

Read the load of the gauge.

Load		
n		lbf
409.2	MIN	91.99
480.4	MAX	107.99

Assemble

Two-Piece Filter Head, Spin-on Filter

The QSK45 engine has three combination filters.

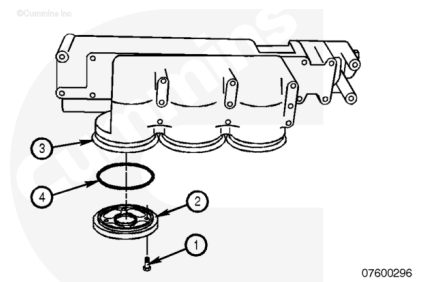
The QSK60 engine has four combination filters.

Oil filter heads that use the combination oil filter have an adapter for the filter.

Lubricate the o-ring (4), and place it into the filter head (3).

Install the adapter (2). Install the capscrews (1).

Tighten the capscrews.



Torque Value: 23 n•m [204 in-lb]

One-Piece Filter Head, Spin-on Filter

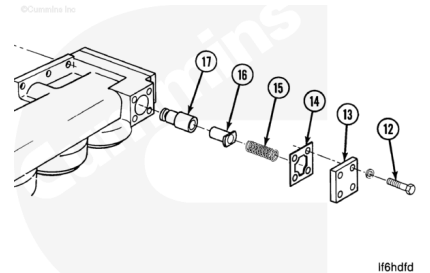
Use clean engine oil. Lubricate the filter bypass valve housing (17) and the plunger (16). Install the filter bypass valve housing and plunger.

Be sure that the plunger moves freely in the housing. If it does **not** move freely, check for debris or burrs on the plunger and assemble again.

Install the:

- Spring (15)
- Gasket (14)
- Cover plate (13)
- Capscrews (12).

Tighten capscrews.

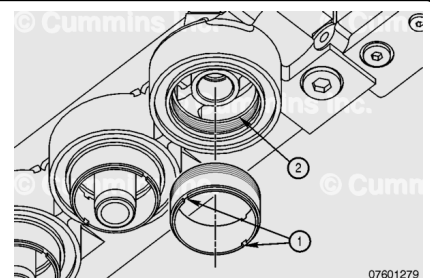


Torque Value: 45 n•m [33 ft-lb]

Apply Loctite® 270, Part Number 3375068, or equivalent, to the adapter threads.

To install the outer adapter, a tool can be fabricated. Insert the fabricated tool into the two notches located in the adapter (1).

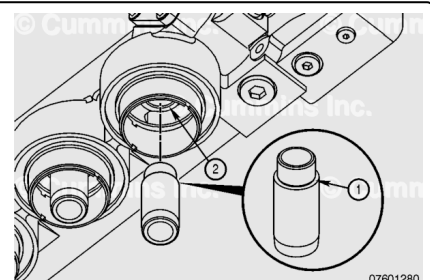
Tighten the adapter until the shoulder touches the casting (2).



Torque Value: 45 n•m [33 ft-lb]

Apply retaining compound, Part Number 3823718, or equivalent, to the inner adapter at (1).

Use a soft mallet to install the inner adapter. Tap the adapter in until the shoulder touches the casting (2).

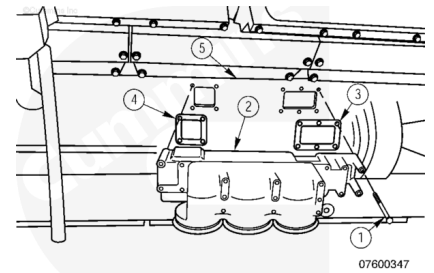


Install

Two-Piece Filter Head, Spin-on Filter

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

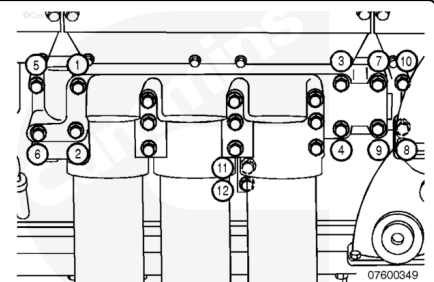


Position the filter head (2) and the gaskets (3 and 4) on the engine (5) with 12 capscrews (1).

Tighten the capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

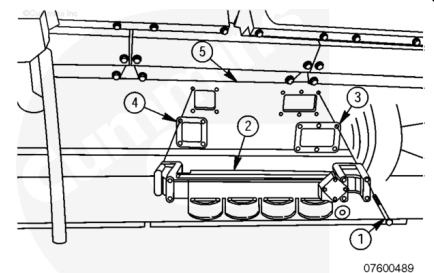
If the engine is equipped with pre-oil filter and post-oil filter sensors, install and connect the sensors.



One-Piece Filter Head, Spin-on Filter

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

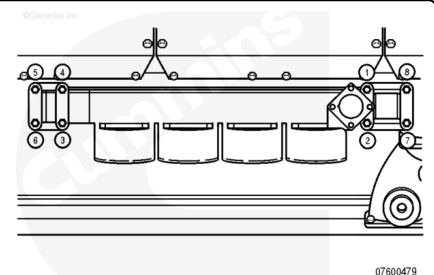


Position the filter head (2) and the gaskets (3 and 4) on the engine (5) with eight capscrews (1).

Tighten the capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]

If the engine is equipped with pre-oil filter and post-oil filter sensors, install and connect the sensors.



Finishing Steps

The QSK45 engine has three combination oil filters.

The QSK60 engine has four combination oil filters.

- Install new combination oil filters. Refer to Procedure 007-002 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-002.html)
- Operate the engine to normal operating temperature. Check for leaks.